

information. Furthermore, we need to analyse the geometry of the image pair without referring to known scene information.

The second task, the reconstruction of unknown scene points, scene lines, or other scene features visible in two images, can be solved by *intersection*, also called *triangulation*. For two corresponding points (χ', χ'') , the determination of the scene point X already is an overconstrained problem, as we have four image coordinates for determining the three spatial coordinates. The accuracy will depend on what is called the *parallactic angle* between the two projection rays.

*intersection,
triangulation*

parallactic angle

The situation is different for straight 3D lines $\mathcal{L}$, not shown in the figure. The two projection planes $\mathcal{A}_l$ and $\mathcal{A}_l''$ spanned by the projection centres and the image lines always intersect. There is *no constraint based on corresponding image lines* on the pose of the cameras as we have four coordinates (parameters) for the observed two image lines, which are necessary for determining the four parameters of the 3D line. If the 3D line is curved, we again will have constraints, but their number depends on the form of the 3D line, e.g., it will be different depending on whether we have a 3D circle, a 3D ellipse, or a free form line in 3D.

*there is no constraint
based on
corresponding image
lines*

The complete procedure for orienting two images is shown in Fig. 13.2. It consists of two

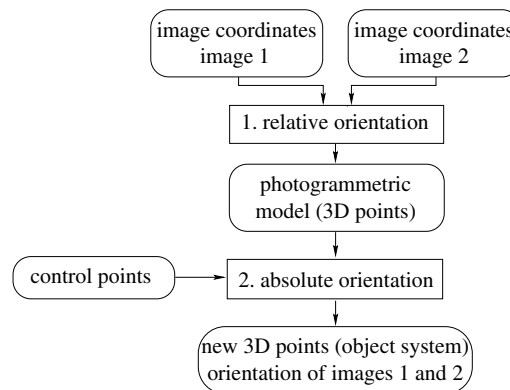


Fig. 13.2 Two-step procedure for the orientation of the image pair

steps. The relative orientation uses only image information, namely corresponding points, and yields a photogrammetric model. It consists of all derived 3D points and the relative pose of the two cameras up to a common similarity or projective transformation into the scene coordinate system. Since the algebraic relations between the image coordinates and the pose parameters are nonlinear and we have to handle possible outliers, we need direct as well as statistically optimal solutions for the relative orientation. The absolute orientation transforms the photogrammetric model into the scene system using control points. The result is statistically suboptimal and can be seen as an approximate solution which may be refined using bundle adjustment as discussed in Chap. 15.

*photogrammetric
model*

13.2 The Geometry of the Image Pair

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The geometry of the image pair describes the geometric relations between the scene points, two cameras and the image points. This is achieved explicitly by the orientation parameters and the scene coordinates or implicitly by constraints, which do not use the coordinates of the scene points. Depending on whether the cameras are calibrated or not, we use the spherical or the perspective camera model (see Sect. 12.7, p. 464) in order to exploit the generality of the algebraic expressions. Thus we use the spherical camera model for calibrated perspective cameras. This simplifies the algebraic description of the geometric relations, collected in what is called epipolar geometry of the image pair.

13.2.1 Number of Parameters for Orientation and Reconstruction

The orientation of an image pair can be described easily by the orientation of the two individual cameras, specifically their exterior and interior orientation, see Sect. 12.1.2.1, p. 460. For calibrated cameras where the IO is known, we need six parameters for the exterior orientation of each image and thus 12 parameters for the image pair.

In the case of uncalibrated cameras we have many options, depending on whether the IOs of the two cameras are the same or not, or whether we only assume the cameras to be straight line-preserving or not. In this chapter, we assume the nonlinear distortions of the two cameras to be sufficiently small enough to be neglected in the application. Furthermore we assume they have different interior orientations. Thus the two cameras follow a straight line-preserving perspective camera model and are characterized by their projection matrices. Then we need 11 parameters for each camera, five for the IO and six for the EO, so we need a total of 22 parameters to describe the orientation of a pair of straight line-preserving cameras.

In both cases, control points or lines, i.e., points or lines known in object space and visible in the images, are required in order to determine the exterior orientation of the cameras.

To find the minimum number of control points required to determine the exterior orientation, we investigate how many and which parameters are already determinable if we have only two images and *no 3D information*.

13.2.1.1 Two Calibrated Cameras

We start with two calibrated cameras. If we had a perfect orientation of the cameras and if the image measurements were perfect, the two corresponding camera rays,² $\mathcal{L}_{x'} = O' \wedge \chi'$ and $\mathcal{L}_{x''} = O'' \wedge \chi''$, would intersect in the scene point X . This is the *coplanarity constraint*,

$$\mathcal{L}_{x'}(\chi', O') \cap \mathcal{L}_{x''}(\chi'', O'') = X, \quad (13.1)$$

since corresponding rays $\mathcal{L}_{x'}$ and $\mathcal{L}_{x''}$ of an oriented image pair are coplanar (see Fig. 13.1). Given the orientation parameters of the images, the 3D point X generally can be derived from its images χ' and χ'' by triangulation. The angle between the two rays $\mathcal{L}_{x'}$ and $\mathcal{L}_{x''}$ is the *parallactic angle* α . The two rays are parallel, i.e., the parallactic angle is zero if the scene point is at infinity, if the two projection centres are identical, or if the ray passes through the two projection centres. In all three cases only the direction to the scene point can be derived. In the following we assume that the two projection centres are distinct if not stated otherwise.

Observe, we did not refer to a specific camera model. For the spherical model the rays, e.g., $\mathcal{L}_{x'} = O' \wedge \chi'$, point from the projection centre to the 3D point. For perspective cameras, the direction of $\mathcal{L}_{x'}$ depends on whether the image is in viewing or taking position (see Sect. 12.1.1, p. 456).

² Interpreting the image points as points in 3D.

*in this chapter we
assume cameras to
be straight
line-preserving*

*coplanarity
constraint*

The two bundles of corresponding rays $\{\mathcal{L}_{x'_i}, \mathcal{L}_{x''_i}\}$ intersect in a set X_i of 3D points. They represent an angle-preserving model of the scene, a photogrammetric model. Nothing more can be derived from these bundles. Especially, the scale of the photogrammetric model cannot be determined, as there is no possibility of deriving length information from the angles encoded in the bundle of imaging rays.

Therefore, the orientation of the two cameras and hence the object can be reconstructed up to a spatial similarity transformation, which is specified by seven parameters (three translation parameters, three rotation parameters, one scale parameter), which means that only $12 - 7 = 5$ parameters out of the 12 total parameters of the exterior orientation of the calibrated image pair are determinable if no object information is available.

We can identify these five parameters: Assume two cameras in arbitrary pose in space which observe a 3D object represented by a set of points. Corresponding image rays intersect in the corresponding space points. What motions of the two cameras are possible if we fix the image points x'_i and x''_i and still require corresponding rays to intersect? The pose in space of the camera pair as a whole may be chosen arbitrarily if the cameras remain in the same relative position. This allows us to fix the pose of the camera pair by fixing the pose of one camera. Corresponding rays will no longer intersect if the second camera is rotated or shifted across the line joining the two cameras.

Only if the second camera is shifted in the direction towards or away from the first camera will corresponding rays remain coplanar. This leads to a geometric scaling, reduction or enlargement, of the point set produced by the intersecting rays (see Fig. 13.3). There is no constraint for corresponding image lines l' and l'' , as discussed above.

relative orientation

Therefore the following parameters fix the *relative orientation* of the two images in terms of the camera parameters:

- The rotation R_{12} of the second camera relative to the first, which involves three parameters.
- The direction B of the base line $O'O''$ connecting the two projection centres O' and O'' . This involves additional two parameters, since the distance of the two projection centres cannot be determined from the coplanarity constraint.

Given enough corresponding image points, the relative orientation is only unique if we assume that the projection rays are oriented, thus if they are half-rays. Otherwise we could reverse the direction of the basis or rotate one camera around the basis by 180° without violating the coplanarity constraint. However, in that case the projection rays would not intersect *in front of the cameras* anymore, a notion which needs to be clarified for spherical cameras. We will discuss this situation in more detail when estimating the relative orientation of two images.

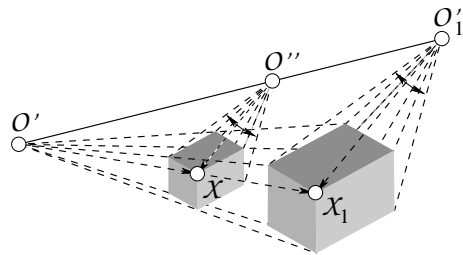


Fig. 13.3 The coplanarity constraint is invariant with respect to a translation of the camera along the base line $O'O''$, e.g., from O'' to O''_1 . The two bundles at O'' and O''_1 have rays with the same spatial directions. This visualizes the fact that the scale of the photogrammetric model cannot be determined from image information alone

Table 13.1 sums up the results so far: The relative orientation of two images taken with calibrated cameras is characterized by five independent parameters. An object can

be reconstructed from two images of calibrated cameras only up to a spatial similarity transformation. The result is a photogrammetric model.

The orientation of the photogrammetric model in space is called the *absolute orientation*. The notion again refers to the process of orientation as well as to the result of the orientation. To determine the absolute orientation of a photogrammetric model derived from images taken with calibrated cameras, at least seven constraints are required.

Table 13.1 Number of free parameters of the orientation $O=IO+EO$, the relative orientation RO of the images, the absolute orientation AO of the photogrammetric model

cameras	# O/image	# O/image pair	# RO	# AO
calibrated spherical perspective	6	12	5	7
uncalibrated perspective	11	22	7	15

13.2.1.2 Two Uncalibrated Cameras

For uncalibrated cameras, which we assume to be straight line-preserving perspective cameras, we make a similar argument. Starting from an oriented pair of uncalibrated cameras we can again reconstruct the scene points from their corresponding image points by triangulation. Since the mapping from scene to image space is straight line-preserving, straight lines, not angles, are the most general invariants in object space. From the information in an image pair we can therefore reconstruct the object up to a straight line preserving transformation (Faugeras, 1992). This is a spatial homography. It is described by a homogeneous 4×4 matrix, thus by a minimum of 15 parameters in the general case. Therefore, only $22 - 15 = 7$ parameters of the interior and the exterior orientation of the image pair can be reconstructed from two images alone. Here the interpretation is less intuitive. However, it can be shown that, along with the rotation and the direction of the base line, we can determine the two principal distances (Hartley, 1992) in a general situation.

Table 13.1 again summarizes the result: The relative orientation of two images of uncalibrated cameras is characterized by seven independent parameters. An object can be reconstructed only up to a spatial homography. The result is also called a photogrammetric model.

The *absolute orientation* of a photogrammetric model of *uncalibrated* straight line-preserving perspective cameras needs at least 15 constraints.

13.2.2 Coplanarity Constraints for Images of Uncalibrated Cameras

We now give explicit expressions for the coplanarity constraint for the relative orientation of two images taken with uncalibrated straight line-preserving cameras which for reasons of generality are assumed to have different interior orientation.

13.2.2.1 The Coplanarity Constraint

Let the two cameras be characterized by the two projection matrices P' and P'' . Thus we have the following mappings:

$$\mathbf{x}' = P' \mathbf{X} \quad \mathbf{x}'' = P'' \mathbf{X}, \quad (13.2)$$

with the projection matrices, see (12.44), p. 474

$$\mathbf{P}' = [\mathbf{A}'|\mathbf{a}'] = \mathbf{K}'\mathbf{R}'[I_3] - \mathbf{Z}' \quad \mathbf{P}'' = [\mathbf{A}''|\mathbf{a}''] = \mathbf{K}''\mathbf{R}''[I_3] - \mathbf{Z}'' . \quad (13.3)$$

The coplanarity constraint of the three space vectors $\mathcal{O}'\mathcal{X}'$, $\mathcal{O}''\mathcal{X}''$ and $\mathcal{O}'\mathcal{O}''$ can be expressed as

$$\det[\mathcal{O}'\mathcal{X}' \quad \mathcal{O}'\mathcal{O}'' \quad \mathcal{O}''\mathcal{X}''] = 0 , \quad (13.4)$$

where $\det[\cdot, \cdot, \cdot]$ is the volume of the parallelepiped of three vectors.

The directions of the vectors $\mathcal{O}'\mathcal{X}'$ and $\mathcal{O}''\mathcal{X}''$ can be derived from the image coordinates $\mathbf{x}'$ and $\mathbf{x}''$ using the parameters of the interior and the exterior orientation, while the *base vector* $\mathcal{O}'\mathcal{O}''$ directly results from the coordinates of the projection centres. The determination of the volume requires that the three vectors are given in mutually parallel coordinate systems. We express the base vector in the scene coordinate system $S := S_o$ and the directions in the normalized camera systems S_n (see the normalized camera in Table 12.1, p. 479),

$$\mathbf{b} = \mathbf{B} = \mathbf{Z}'' - \mathbf{Z}' , \quad {}^n\mathbf{x}' = \mathbf{R}'^T(\mathbf{K}')^{-1}\mathbf{x}' , \quad {}^n\mathbf{x}'' = \mathbf{R}''^T(\mathbf{K}'')^{-1}\mathbf{x}'' ; \quad (13.5)$$

the coplanarity constraint (13.4) is

$$|{}^n\mathbf{x}' , \mathbf{b} , {}^n\mathbf{x}''| = \begin{vmatrix} {}^n x' & B_x & {}^n x'' \\ {}^n y' & B_y & {}^n y'' \\ 1 & B_z & 1 \end{vmatrix} = {}^n\mathbf{x}' \cdot (\mathbf{b} \times {}^n\mathbf{x}'') = {}^n\mathbf{x}'^T \mathbf{S}_b {}^n\mathbf{x}'' = 0 . \quad (13.6)$$

Using (13.5) we have, explicitly,

$$\mathbf{x}'^T (\mathbf{K}')^{-T} \mathbf{R}'^T \mathbf{S}_b \mathbf{R}''^T (\mathbf{K}'')^{-1} \mathbf{x}'' = 0 . \quad (13.7)$$

The constraint is a bilinear form in the vectors $\mathbf{x}'$ and $\mathbf{x}''$ and depends on the ten parameters of the two calibration matrices, the six rotation parameters and the two parameters of the direction of the basis. However, the relative orientation of two images of uncalibrated cameras has only seven degrees of freedom, a difference which will be relevant when deriving the projection matrices from a given fundamental matrix in Sect. 13.3.7.1, p. 594.

13.2.2.2 The Fundamental Matrix

We therefore analyse the 3×3 matrix

$$\mathbf{F} = (\mathbf{K}')^{-T} \mathbf{R}'^T \mathbf{S}_b \mathbf{R}''^T (\mathbf{K}'')^{-1} \quad (13.8)$$

of the bilinear form (13.7). It is called the *fundamental matrix* of the relative orientation of a pair of images of uncalibrated cameras. The general form of the coplanarity constraint then reads as

$$\mathbf{x}'^T \mathbf{F} \mathbf{x}'' = 0 . \quad (13.9)$$

*coplanarity
constraint for
uncalibrated
cameras*

Several remarks are useful here:

- The coplanarity constraint is bilinear in the homogeneous image coordinates $\mathbf{x}'$ and $\mathbf{x}''$ and linear in the elements of the fundamental matrix. This is the basis for a simple determination of the fundamental matrix from corresponding points.
- The fundamental $\mathbf{F}$ matrix has seven degrees of freedom. This is because $\mathbf{F}$ is homogeneous and singular, since the skew symmetric matrix $\mathbf{S}_b$ is singular with rank 2. Therefore any matrix of the form

$$\mathbf{F} = \mathbf{U} \text{Diag}([s_1, s_2, 0]) \mathbf{V}^T , \quad s_i > 0 , \quad (13.10)$$

with orthogonal matrices U and V , is a fundamental matrix. As shown in Sect. 13.2.1.2, p. 552, seven parameters are *necessary* for describing the relative orientation of two images taken with uncalibrated cameras. Since the fundamental matrix is *sufficient* for describing the relative orientation in terms of the epipolar geometry and has seven degrees of freedom, the fundamental matrix contains the complete information about the relative orientation of two images of uncalibrated cameras. Since we assume a central projection, the seven parameters refer to five parameters for the relative pose of the two cameras and to two parameters of the interior orientation of the two cameras.

*fundamental matrix
from projection
matrices*

- If the projection matrices are given, we can derive the fundamental matrix without the complete partitioning of P' and P'' . Let the projection matrices be partitioned into a left 3×3 matrix and a 3-vector as in (13.3), then the fundamental matrix is given by

$$F = A'^{-T} S_{b_{12}} A''^{-1} \quad \text{with} \quad b_{12} = A''^{-1} a'' - A'^{-1} a'. \quad (13.11)$$

This is because the projection centres are $Z' = -A'^{-1} a'$ and $Z'' = -A''^{-1} a''$, $A' = K'R'$ and $A'' = K''R''$. Equation (13.11) can also be written as

$$F = A'^{\circ} S_{b_{12}} A''^{\circ T}, \quad (13.12)$$

since for any square matrix A we have $A^{-1} = A^{\circ T}/|A|$ with the cofactor matrix A° , see (A.19), p. 769. If the first projection matrix is fixed to be $P' = [I_3 | 0]$, and $P'' = [A'' | a'']$, the fundamental matrix has the form

$$F = A'' S(a''), \quad (13.13)$$

since $F = S_{b_{12}} A''^{\circ T} = S_{A''^{\circ T} a''} A''^{\circ T} = A'' S(a'')$ due to $b_{12} = A''^{-1} a = A''^{\circ T} a$ and (A.47), p. 772 and (A.25), p. 770.

*fundamental matrix
from camera planes*

- An alternative expression for the fundamental matrix F exploits the tools from algebraic projective geometry, which will be used when describing the relations between three images.

The derivation starts from the projection matrices of the two cameras,

$$P_1 := P' = \begin{bmatrix} A_1^T \\ B_1^T \\ C_1^T \end{bmatrix} \quad P_2 := P'' = \begin{bmatrix} A_2^T \\ B_2^T \\ C_2^T \end{bmatrix}, \quad (13.14)$$

the vectors A_i , etc., which are 4-vectors representing the camera planes of each camera, i.e., the planes passing through the projection centre and the axes of the sensor coordinate system (see Sect. 12.1.3.8, p. 473).

The fundamental matrix is given by the 3×3 matrix

$$F = - \begin{bmatrix} |B_1, C_1, B_2, C_2| & |B_1, C_1, C_2, A_2| & |B_1, C_1, A_2, B_2| \\ |C_1, A_1, B_2, C_2| & |C_1, A_1, C_2, A_2| & |C_1, A_1, A_2, B_2| \\ |A_1, B_1, B_2, C_2| & |A_1, B_1, C_2, A_2| & |A_1, B_1, A_2, B_2| \end{bmatrix}, \quad (13.15)$$

composed of 4×4 determinants. It results from the coplanarity constraint that the two projection lines $L_{x'}$ and $L_{x''}$ intersect,

$$L_{x'}^T \bar{L}_{x''} = 0. \quad (13.16)$$

We can see this when using (12.85), p. 483

$$L_{x'} = u'(B_1 \cap C_1) + v'(C_1 \cap A_1) + w'(A_1 \cap B_1) \quad (13.17)$$

$$L_{x''} = u''(B_2 \cap C_2) + v''(C_2 \cap A_2) + w''(A_2 \cap B_2) \quad (13.18)$$

and collecting the coefficients for products $u'u''$, etc., e.g., $F_{11} = (\mathbf{B}_1 \cap \mathbf{C}_1)^\top (\mathbf{B}_2 \cap \mathbf{C}_2) = -[\mathbf{B}_1, \mathbf{C}_1, \mathbf{B}_2, \mathbf{C}_2]$, due to (7.61), p. 304. With the rows $\bar{\mathbf{L}}_{1i}$ and $\bar{\mathbf{L}}_{2j}$ of the projection matrices $\mathbf{Q}'$ and $\mathbf{Q}''$, see (12.77), p. 481, this is equal to $F_{11} = \mathbf{L}_{11}^\top \bar{\mathbf{L}}_{21}$. Therefore the fundamental matrix can also be written as

$$\mathbf{F} = [F_{ij}] = [\mathbf{L}_{1i}^\top \bar{\mathbf{L}}_{2j}] = \mathbf{Q}' \bar{\mathbf{Q}}''^\top, \quad (13.19)$$

a form we will use when discussing the geometry of the image pair in Sect. 13.2.5, p. 562.

- If the fundamental matrix is known together with the covariance matrix of its elements (see Sect. 13.3.2.2, p. 572) we can check the correspondence of two points $\mathbf{x}'$ and $\mathbf{x}''$. Using the Kronecker product (see Sect. A.7, p. 775) the left side of the coplanarity constraint (13.9) will yield a residual

$$w = (\mathbf{x}'' \otimes \mathbf{x}')^\top \mathbf{f}, \quad (13.20)$$

with the vector $\mathbf{f} = \text{vec} \mathbf{F}$ collecting the columns of $\mathbf{F}$. The variance of w can be given explicitly as

$$\sigma_w^2 = [\mathbf{l}'^\top \mid \mathbf{l}''^\top] \begin{bmatrix} \Sigma_{\mathbf{x}'\mathbf{x}'} & \Sigma_{\mathbf{x}'\mathbf{x}''} \\ \Sigma_{\mathbf{x}''\mathbf{x}'} & \Sigma_{\mathbf{x}''\mathbf{x}''} \end{bmatrix} \begin{bmatrix} \mathbf{l}' \\ \mathbf{l}'' \end{bmatrix} \quad (13.21)$$

$$+ (\mathbf{x}'' \otimes \mathbf{x}')^\top \Sigma_{\mathbf{F}} (\mathbf{x}'' \otimes \mathbf{x}'), \quad (13.22)$$

since we have the Jacobians

$$\frac{\partial w}{\partial \mathbf{x}'} = \mathbf{x}''^\top \mathbf{F}^\top =: \mathbf{l}'^\top, \quad \frac{\partial w}{\partial \mathbf{x}''} = \mathbf{x}'^\top \mathbf{F} =: \mathbf{l}''^\top, \quad \frac{\partial w}{\partial \mathbf{f}} = (\mathbf{x}'' \otimes \mathbf{x}')^\top. \quad (13.23)$$

Here we assumed that the observed coordinates $\mathbf{x}'$ and $\mathbf{x}''$ are correlated, see the discussion in Sect. 10.6.3.1, p. 425 and exercise 21, p. 435. In addition we assumed the coordinates are independent of the elements of the fundamental matrix. This will certainly be the case if the point pair has not taken part in the determination of the fundamental matrix. The test statistic

*statistical test for
correspondence*

$$z = \frac{w}{\sigma_w} \sim \mathcal{N}(0, 1) \quad (13.24)$$

can be tested for significance. It is normally distributed if the perspective model holds and the two points are corresponding. Thus we will reject the hypothesis that the two points $\mathbf{x}'$ and $\mathbf{x}''$ correspond if

$$|z| > k_\alpha, \quad (13.25)$$

where k_α is the $(1-\alpha)$ percentile of the normal distribution, e.g., $k_\alpha = 1.96$ for $\alpha = 5\%$.

- The definition (13.8), p. 553 of the fundamental matrix is not the same as in Hartley and Zisserman (2000) and Faugeras and Luong (2001). It generally differs just by a transposition. However, in the context of many images we will call $\mathbf{F}_{ij}$ that fundamental matrix which yields the constraint $\mathbf{x}_i'^\top \mathbf{F}_{ij} \mathbf{x}_j' = 0$. Thus in our case, for images 1 and 2, we have $\mathbf{F} \doteq \mathbf{F}_{12}$.³

13.2.3 Coplanarity Constraint for Images of Calibrated Cameras

For images of calibrated cameras the coplanarity constraint can be simplified by using the directions ${}^c\mathbf{x}'$ and ${}^c\mathbf{x}''$ of the camera rays of a *spherical camera*. If we start from sensor coordinates $\mathbf{x}'$ and $\mathbf{x}''$ of a calibrated perspective camera, the directions ${}^c\mathbf{x}'$ and ${}^c\mathbf{x}''$ from

³ The definition in Hartley and Zisserman (2000) is $\mathbf{F} \doteq \mathbf{F}_{21} = \mathbf{F}_{12}^\top$, due to $\mathbf{x}_i''^\top \mathbf{F} \mathbf{x}_i' = 0$.

the projection centres to the image points in the camera coordinate systems $S_{c'}$ and $S_{c''}$ are

$${}^c\mathbf{x}' = -\text{sign}(c')K'^{-1}\mathbf{x}' = -\text{sign}(c') \begin{bmatrix} {}^ix' \\ {}^iy' \\ c' \end{bmatrix} \quad (13.26)$$

(see (12.22), p. 469 and (12.109), p. 492) and similarly for ${}^c\mathbf{x}''$.⁴

Hence, in the case of calibrated perspective cameras with principal distances c' and c'' , which need not be identical, we obtain the elements of the direction vectors explicitly from the image coordinates ${}^ix'$ and ${}^ix''$ in the ideal cameras.

In the following we allow for bundles of rays of two *calibrated perspective cameras* when determining the relative orientation. The cameras then also may be *calibrated spherical cameras*, e.g., if the camera rays are derived from the image coordinates of a fish-eye or a catadioptric camera.

relative orientation
of images of
spherical cameras

13.2.3.1 The Essential Matrix

From the coplanarity constraint (13.7) for uncalibrated cameras we immediately obtain

$${}^c\mathbf{x}'^T R' S_b R''^T {}^c\mathbf{x}'' = 0, \quad (13.27)$$

which is equivalent to requiring the determinant

$$|R'^T {}^c\mathbf{x}', \mathbf{b}, R''^T {}^c\mathbf{x}''| = 0$$

to vanish. We define the *essential matrix*,

$$\mathbf{E} \doteq R' S_b R''^T, \quad (13.28)$$

and obtain the coplanarity constraint for calibrated cameras,

$${}^c\mathbf{x}'^T \mathbf{E} {}^c\mathbf{x}'' = 0. \quad (13.29)$$

With observed image coordinates ${}^ix'$ and ${}^ix''$ in an ideal camera, see (13.26), p. 556, we explicitly have

$$[{}^ix', {}^iy', c'] \begin{bmatrix} e_1 & e_4 & e_7 \\ e_2 & e_5 & e_8 \\ e_3 & e_6 & e_9 \end{bmatrix} \begin{bmatrix} {}^ix'' \\ {}^iy'' \\ c'' \end{bmatrix} = 0 \quad (13.30)$$

independent of the sign of the principal distances. Some remarks are helpful:

- The coplanarity constraint for calibrated cameras is a bilinear form in the direction vectors ${}^c\mathbf{x}'$ and ${}^c\mathbf{x}''$ and linear in the elements of the essential matrix $\mathbf{E}$. This again gives rise to direct solutions for the essential matrix.
- The essential matrix $\mathbf{E}$ has five degrees of freedom. Therefore it has to fulfil $9 - 5 = 4$ constraints which can be expressed as a function of the columns of $\mathbf{E}$ and its cofactor matrix $\mathbf{E}^\circ = \text{cof}(\mathbf{E})$ (A.19), p. 769,

$$\mathbf{E} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3] \quad \mathbf{E}^\circ = [\mathbf{a}_2 \times \mathbf{a}_3, \mathbf{a}_3 \times \mathbf{a}_1, \mathbf{a}_1 \times \mathbf{a}_2] = [\mathbf{a}_1^\circ, \mathbf{a}_2^\circ, \mathbf{a}_3^\circ] \quad (13.31)$$

It is homogeneous. Therefore we may fix its squared Frobenius norm to 2,

$$\|\mathbf{E}\|^2 = |\mathbf{a}_1|^2 + |\mathbf{a}_2|^2 + |\mathbf{a}_3|^2 = 2, \quad (13.32)$$

and this fixes the length of the base vector to $|\mathbf{b}| = 1$. Moreover we have the constraints (Rinner, 1963)

constraints on the
essential matrix

⁴ Assuming $|K| > 0$ and $\mathbf{x}'$ positive, thus $x_h > 0$.

$$\begin{aligned}
|\mathbf{a}_1|^2 + |\mathbf{a}_1^\circ|^2 &= 1 \\
|\mathbf{a}_2|^2 + |\mathbf{a}_2^\circ|^2 &= 1 \\
|\mathbf{a}_3|^2 + |\mathbf{a}_3^\circ|^2 &= 1.
\end{aligned} \tag{13.33}$$

Proof: Without loss of generality we may choose $\mathbf{E} = \mathbf{S}(\mathbf{b})$, since we may replace $R^\top \mathbf{a}_i = \mathbf{c}_i$, $R^\top(\mathbf{a}_i \times \mathbf{a}_j) = \mathbf{c}_i \times \mathbf{c}_j$ and $\mathbf{c}_i^\circ = R^\top \mathbf{a}_i^\circ$. If we assume $|\mathbf{b}| = |(a, b, c)^\top| = 1$, we get the proof for index 1:

$$|\mathbf{c}_1|^2 + |\mathbf{c}_1^\circ|^2 = \left| \begin{bmatrix} 0 \\ c \\ -b \end{bmatrix} \right|^2 + \left| \begin{bmatrix} -c \\ 0 \\ a \end{bmatrix} \times \begin{bmatrix} b \\ -a \\ 0 \end{bmatrix} \right|^2 = (b^2 + c^2) + a^2 \left| \begin{bmatrix} a \\ b \\ c \end{bmatrix} \right|^2 = 1.$$

Rinner (1963) shows that the four constraints (13.32) and (13.33) are independent. $\diamond$

- The essential matrix is singular

$$|\mathbf{E}| = 0. \tag{13.34}$$

- We have the nine constraints (see Stefanovic, 1973),

$$\mathbf{E}\mathbf{E}^\top \mathbf{E} - \frac{1}{2} \text{tr}(\mathbf{E}\mathbf{E}^\top) \mathbf{E} = \mathbf{0} \tag{13.35}$$

which result from the cubic of a skew 3×3 matrix (A.4.2), p. 771, and will be used for a direct solution for relative orientation with five points in Sect. 13.3.2.4, p. 575.

- Finally we have the SVD of an essential matrix,

$$\mathbf{E} = \mathbf{U} \text{Diag}([s, s, 0]) \mathbf{V}^\top, \quad s > 0, \tag{13.36}$$

SVD of essential matrix

where the first two singular values are identical. The SVD is not unique since with any rotation matrix $R = R_3(\alpha)$ we have $\mathbf{E} = (\mathbf{U}R) \text{Diag}([s, s, 0]) (R^\top \mathbf{V}^\top) = \bar{\mathbf{U}} \text{Diag}([s, s, 0]) \bar{\mathbf{V}}^\top$. This freedom can be used to specify one of the two orthogonal matrices, say $\mathbf{U}$, by only two parameters. Due to the homogeneity of the essential matrix it therefore has five degrees of freedom.

13.2.3.2 Parametrizations of Relative Orientation

We now take into account that the relative orientation of two images is determined by only five independent parameters and give three classical parametrizations which have an intuitive geometric interpretation. We refer you to Fig. 13.4, p. 557.

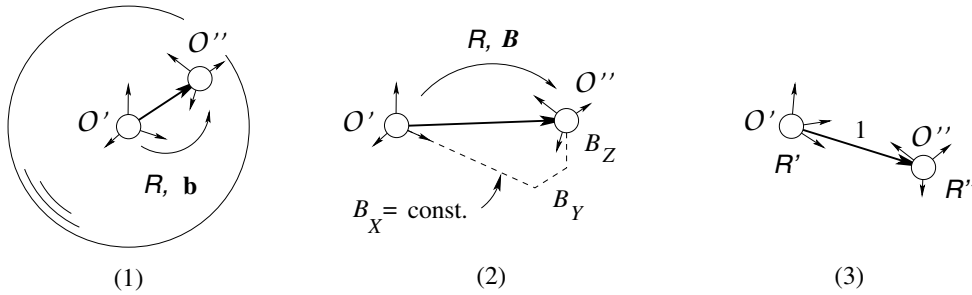


Fig. 13.4 Parametrization of relative orientation of two images taken with calibrated cameras. (1) General parametrization of dependent images with the normalized direction vector $\mathbf{b}$, with $|\mathbf{b}| = 1$ and the rotation matrix R . Thus the second projection centre O'' is located on the unit sphere around the first projection centre O . (2) Photogrammetric parametrization of dependent images with two components B_Y and B_Z of the base vector $\mathbf{B}$ and the rotation matrix R . The component B_X of the base vector is fixed, either to 1 or to another value close to the length of the basis in the scene coordinate system. (3) Parametrization with independent images using two rotation matrices, $R'(\phi', \kappa')$ and $R''(\omega'', \phi'', \kappa'')$. Here the base vector defines the X-axis of the image pair

Parametrization for Dependent Images. Starting from (13.28), p. 556 we refer to the coordinate system of the first camera and describe the relative orientation via the direction of the base vector and the rotation parameters of the second camera: Thus we assume $R' = I_3$. The mutual rotation $R := R''$ and the basis $\mathbf{b} = \mathbf{B}$ refer to the coordinate system of the first camera. The coordinate systems of the base vector and the rotation matrices are not indicated.

With the rotation matrix R and the skew symmetric matrix of the base vector $\mathbf{b} = \mathbf{B}$, the essential matrix is

$$\mathbf{E} = \mathbf{S}_b R^T \quad (13.37)$$

and the coplanarity constraint reads

$${}^c\mathbf{x}'^T \mathbf{S}_b R^T {}^c\mathbf{x}'' = 0. \quad (13.38)$$

Since the base vector contains three elements but has only two degrees of freedom, since the length is arbitrary, we need to impose one constraint.

This can be done in two ways:

*general
parametrization with
dependent images*

1. General parametrization: We describe the relative pose of the second camera using the direction $\mathbf{b}/|\mathbf{b}|$ of the base vector $\mathbf{b} = [B_X, B_Y, B_Z]^T$ and the rotation matrix R . We therefore obtain

$${}^c\mathbf{x}'^T \mathbf{S}_b R^T {}^c\mathbf{x}'' = 0 \quad \text{with} \quad |\mathbf{b}| = 1. \quad (13.39)$$

Thus the relative orientation is described by six parameters and one additional constraint on the length of the base vector $\mathbf{b}$:

$$(B_X, B_Y, B_Z, \omega, \phi, \kappa) \quad \text{with} \quad B_X^2 + B_Y^2 + B_Z^2 = 1, \quad (13.40)$$

all elements referring to the coordinate system of the first camera.

*photogrammetric
parametrization
with
dependent images*

2. Classical photogrammetric parametrization: Here we assume the direction of the base vector to be approximately in the X -direction; for aerial images very often this is the flight direction. We may then arbitrarily fix the length of the base vector in this direction and obtain

$${}^c\mathbf{x}'^T \mathbf{S}_b R^T {}^c\mathbf{x}'' = 0 \quad \text{with} \quad B_X = \text{constant}. \quad (13.41)$$

Thus we parametrize the relative orientation with the three parameters of the rotation matrix R and the two components B_Y and B_Z of the base vector,

$$(B_Y, B_Z, \omega, \phi, \kappa), \quad (13.42)$$

again all elements referring to the coordinate system of the first camera.

In all cases the basis $\mathbf{b}$ is identical to the left epipole ${}^c\mathbf{e}'$, measured in the first camera system,

$$\mathbf{b} = {}^c\mathbf{e}', \quad (13.43)$$

since ${}^c\mathbf{e}'^T \mathbf{E} = \mathbf{b}^T \mathbf{S}(\mathbf{b}) R^T = 0$.

Both parametrizations for dependent images are asymmetric with respect to the two images. Therefore we sometimes use the following symmetric parametrization, e.g., when rectifying images into what is called the normal case, which is useful for stereo analysis, see Sect. 13.2.6, p. 565.

*parametrization with
independent images*

Parameterization with Independent Images. Here we fix the base vector $\mathbf{B}$ and describe the relative orientation with parameters of independent rotations, R'_i and R''_i , for the cameras, the index standing for independent images.

The base vector is assumed to be $\mathbf{b} = [B_X, 0, 0]^T$ with a priori fixed length $B_X = \text{const.}$ The rotations of both cameras together need six parameters. A common rotation of the two cameras around the X -axis is not determinable from the two image bundles; it would lead to a rotation of the camera pair as a whole, which is part of the absolute orientation. Therefore the rotation angles are constrained such that only the difference $\Delta\omega = \omega'' - \omega'$ of the two cameras, or just one ω -rotation around the X axis, usually ω'' , is a free parameter.

We therefore obtain from (13.28), p. 556,

$${}^c\mathbf{x}'^T R' S_b R''^T {}^c\mathbf{x}'' = 0 \quad \text{with constant } S_b, \quad \omega' = -\omega'' = -\frac{1}{2} \Delta\omega, \quad (13.44)$$

with the five parameters

$$(\Delta\omega, \phi', \kappa', \phi'', \kappa''), \quad (13.45)$$

which are symmetric w.r.t. the two cameras.

Discussion of the Parametrizations:

- All of the mentioned parametrizations demonstrate that the relative orientation can be *parametrized with angles only*, since even in the parametrization (2), see (13.41), p. 558, the direction of the base vector is parametrized via the direction cosines B_Y/B_X and B_Z/B_X . This is to be expected since the points of an image together with the projection centre geometrically represent two *bundles of rays* from which only angles can be derived, not distances.
- While these two parametrizations are general and thus can represent all two-camera configurations, the classical photogrammetric parametrization (2), see (13.41), p. 558, has a singularity in the following situation: If the base vector is directed orthogonal to the X axis, the base components B_Y and B_Z generally will be infinitely large. This parametrization therefore leads to instabilities close to such a two-camera configuration.
- In all cases one base component is fixed to be nonzero, defining the scale of the photogrammetric model.
- The coordinate system chosen for relative orientation is identical to the coordinate system of the photogrammetric model.
- For normalized cameras (see Fig. 12.7, p. 464) with $R' = R'' = I_3$ we obtain the essential matrix

$$\mathbf{E} = S_b. \quad (13.46)$$

As the general parametrization for dependent images (13.39) and (13.40), p. 558 is the simplest one, we will use it in the following and establish the relations to the classical photogrammetric parametrization where appropriate.

Camera Poses Consistent with the Relative Orientation The projection matrices for both images can easily be given in the local coordinate system chosen for the specific parametrizations, namely

$$\mathbf{P}'_d = \mathbf{K}'[I_3|\mathbf{0}] \quad \text{or} \quad \mathbf{P}'_i = \mathbf{K}'R'_i[I_3|\mathbf{0}] \quad (13.47)$$

$$\mathbf{P}''_d = \mathbf{K}''R_d[I_3|-\mathbf{B}_d] \quad \text{or} \quad \mathbf{P}''_i = \mathbf{K}''R''_i[I_3|-\mathbf{B}_i], \quad (13.48)$$

where the indices d and i of the rotation matrices and the base vectors refer to the parametrization with *dependent* and *independent* images.

If we know the projection matrix

$$\mathbf{P}' = \mathbf{K}'R'[I_3|-\mathbf{Z}'] \quad (13.49)$$

*singularity of the
classical
photogrammetric
parametrization*

of the first camera referring to the scene coordinate system, we can derive the projection matrix $P'' = K''R''[I_3 | -Z'']$ for the second camera referring to the scene coordinate system if the relative orientation is given, provided we fix the scale of the base vector in the scene coordinate system and assume the second calibration matrix K'' to be known. Thus we want to determine R'' and Z'' from the parameters of the relative orientation. In both cases we perform an adequate motion of the image pair. In the following we assume the base vectors B_d and B_i have the desired scene scale.

1. For the representation with dependent images we use the rotation matrix R_d and the base vector B_d from the relative orientation.

The projection matrix P' referring to the scene coordinate system is related to the projection matrix P'_d referring to the first image by

$$P' = K'R'[I_3 | -Z'] = \underbrace{K'[I_3 | 0]}_{P'_d} \begin{bmatrix} R'^T & Z' \\ \mathbf{0}^T & 1 \end{bmatrix}^{-1} = P'_d \begin{bmatrix} R' & -R'Z' \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (13.50)$$

Applying the same motion, we obtain the projection matrix for the second image,

$$P'' = \underbrace{K''R_d[I_3 | -B_d]}_{P''_d} \begin{bmatrix} R' & -R'Z' \\ \mathbf{0}^T & 1 \end{bmatrix} = K'' \underbrace{R_d R'}_{R''} [I_3 | -(\underbrace{Z' + R'^T B_d}_{Z''})]. \quad (13.51)$$

Thus we have the relations

$$R'' = R_d R' \quad \text{and} \quad Z'' = Z' + R'^T B_d, \quad (13.52)$$

which are plausible as $R_d = R''R'^T$ and $B_d = R'^T(Z'' - Z')$.

2. For the representation with independent images we use the rotation matrices R'_i and R''_i and the base vector B_i from the relative orientation.

Here the projection matrix P' referring to the scene coordinate system is related to the projection matrix P'_i in the system of the photogrammetric model by a motion where the rotation is the difference $R'^T R'_i$ of the two rotations R' and R'_i . Thus we have

$$P' = K'R'[I_3 | -Z'] = \underbrace{K'R'_i[I_3 | 0]}_{P'_i} \begin{bmatrix} R'^T R'_i & Z' \\ \mathbf{0}^T & 1 \end{bmatrix}^{-1} = P'_i \begin{bmatrix} R'_i{}^T R' & -R'_i{}^T R'Z' \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (13.53)$$

Applying this motion to the second projection, we obtain

$$P'' = \underbrace{K''R''_i[I_3 | -B_i]}_{P''_i} \begin{bmatrix} R'_i{}^T R' & -R'_i{}^T R'Z' \\ \mathbf{0}^T & 1 \end{bmatrix} = K''R''_i R'_i{}^T R' [I_3 | -(\underbrace{Z' + R'^T R'_i B_i}_{Z''})]. \quad (13.54)$$

Thus we have the relations

$$R'' = R''_i R'_i{}^T R' \quad \text{and} \quad Z'' = Z' + R'^T R'_i B_i, \quad (13.55)$$

which are plausible as the rotation differences in the scene system and the system of the photogrammetric model are the same, thus $R''_i R'_i{}^T = R''R'^T$, and the base vector in the photogrammetric system $B_i = R'_i{}^T R'(Z'' - Z')$ is obtained from the base vector $B = Z'' - Z'$ in the scene system by the rotation $R'_i{}^T R'$.

13.2.4 The Normal Case of the Image Pair

Similarly to a single image, we have a *normal case* for the image pair: Both cameras are ideal and identical, the viewing directions are parallel and orthogonal to the base vector, and the x' - and x'' -axes are parallel to the basis.

The normal case is close to the geometric configuration of the human stereo vision system. In photogrammetry, the normal case of the image pair is used for analysing the geometric configuration based on the theoretical accuracy. It can often be used as an approximation for the configuration of a pair of aerial vertical view images.

We first give a geometric derivation of the coplanarity constraint before we derive it algebraically (Fig. 13.5). If the image coordinates of the points χ' and χ'' are error-free,

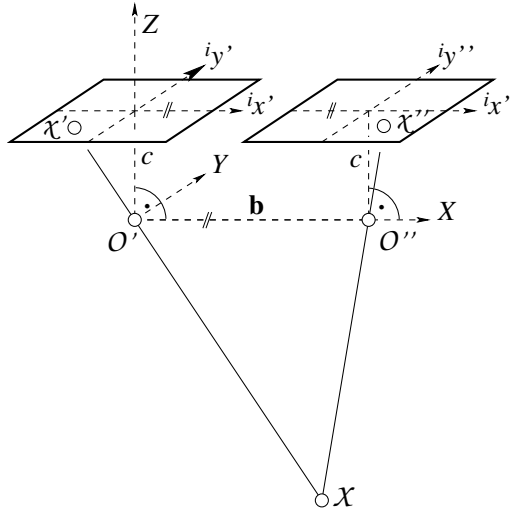


Fig. 13.5 Geometry of the normal case of the image pair. The image points χ' and χ'' , the projection centres O' and O'' and the scene point X are coplanar. The $^i y$ -coordinates of the image points should be the same

the corresponding rays intersect in the object point X . The constraint for the rays to intersect depends only on the $^i y'$ and $^i y''$ coordinates since changes in the $^i x'$ or the $^i x''$ coordinates lead to a change of the distance of the scene point from the base line, while the rays still intersect. The difference

$$p_y = ^i y'' - ^i y' \quad (13.56)$$

is called the *y-parallax*. Obviously the *y-parallax* needs to be zero, meaning that the $^i y$ coordinates of corresponding points have to be identical. This leads to the coplanarity constraint,

$$p_y = ^i y'' - ^i y' = 0. \quad (13.57)$$

When referring to the human visual system, the difference p_y of the y coordinates is often called *vertical parallax*.

The algebraic derivation of the coplanarity constraint starts from the special configuration shown in Fig. 13.5,

$$R' = R'' = I_3 \quad \mathbf{b} = \begin{bmatrix} B_X \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{K} \doteq \mathbf{K}' = \mathbf{K}'' = \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (13.58)$$

i.e., $c = c' = c''$. With

$$\mathbf{E} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -B_X \\ 0 & B_X & 0 \end{bmatrix}, \quad (13.59)$$

we obtain the following coplanarity constraint:

$$[{}^i x' \ {}^i y' \ c] \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -B_X \\ 0 & B_X & 0 \end{bmatrix} \begin{bmatrix} {}^i x'' \\ {}^i y'' \\ c \end{bmatrix} = c B_X ({}^i y'' - {}^i y') = 0. \quad (13.60)$$

The residual of the constraint, if it is not fulfilled, can therefore be used to derive the y -parallax by dividing it by c and B_X . Since neither of them can be zero we finally obtain the coplanarity constraint for the normal case (13.57).

The different versions of the essential matrix are collected in Table 13.2. In addition the table contains the two special cases, namely the normal case and relative orientation with two normalized cameras. In both cases, the number of parameters is less than five, so that they are real specializations. The normal case is relevant for automatic image analysis since all matching procedures can take advantage of the much more simplified equations. The relative orientation of images of normalized cameras occurs when the rotation matrices of two calibrated cameras are measured or determined, e.g., from vanishing points.

Table 13.2 Parametrizations of the coplanarity constraint with the fundamental matrix $\mathbf{F}$ for straight line-preserving uncalibrated cameras and the essential matrix $\mathbf{E}$ for calibrated cameras. Lines 2 and 3: rotation matrix $\mathbf{R}$ and angles ω , ϕ and κ refer to the second camera; $\mathbf{R} \doteq \mathbf{R}''$

camera/parametrization a priori constraints	matrix $\mathbf{F}$ or $\mathbf{E}$	Eqn.	free parameters (#) add. constraints
uncalibrated perspective camera	$\mathbf{F} = \mathbf{K}'^{-\top} \mathbf{R}' \mathbf{S}_b \mathbf{R}''^{\top} \mathbf{K}''^{-1}$	(13.8)	$\mathbf{F}$ (9) $ \mathbf{F} = 1, \mathbf{F} = 0$
spherical camera, calibrated perspective camera			
indep. images general $\mathbf{R}' = \mathbf{I}_3$	$\mathbf{E} = \mathbf{S}_b \mathbf{R}^{\top}$	(13.37)	$\mathbf{b}, \omega, \phi, \kappa,$ (6) $ \mathbf{b} =1$
indep. images special $\mathbf{R}' = \mathbf{I}_3, B_X = \text{const.}$	$\mathbf{E} = \mathbf{S}_b \mathbf{R}^{\top}$	(13.37)	$B_Y, B_Z, \omega, \phi, \kappa$ (5)
dependent images $\mathbf{b} = (B_X, 0, 0)^{\top}, \omega' = -\omega'' = -\frac{1}{2} \Delta\omega$	$\mathbf{E} = \mathbf{R}' \mathbf{S}_b \mathbf{R}''^{\top}$	(13.59)	$\Delta\omega, \phi', \kappa', \phi'', \kappa''$ (5)
normalized cameras $\mathbf{R}' = \mathbf{R}'' = \mathbf{I}_3$ $\mathbf{K}' = \mathbf{K}'' = \mathbf{I}_3$	$\mathbf{E} = \mathbf{S}_b$	(13.46)	$\mathbf{b}$ (3) $ \mathbf{b} = 1$
normal case $\mathbf{R}' = \mathbf{R}'' = \mathbf{I}_3$ $\mathbf{K}' = \mathbf{K}'' = \text{Diag}([c, c, 1])$ $\mathbf{b} = (B_X, 0, 0)^{\top}$	$\mathbf{E} = \mathbf{S}_b$	(13.59)	–

13.2.5 Epipolar Geometry

We now address the problem of predicting the position of a point χ'' in the second image if the point χ' is given in the first image and the relative orientation is known. We will find that the point χ'' lies on a straight line for a perspective camera and on a circle for

a spherical camera. The underlying geometry is the *epipolar geometry* of the image pair. This knowledge can be used (1) for guiding an operator when measuring a point which has already been measured in one other image, and (2) for reducing the search space for finding corresponding image points in automatic image matching.

We define the following entities (Fig. 13.6), which can be used for the perspective and the spherical camera models:

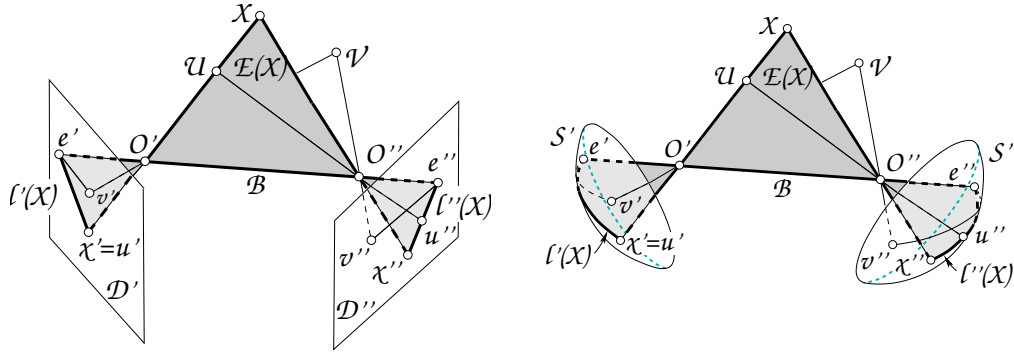


Fig. 13.6 Elements of the epipolar geometry. **Left:** perspective images. **Right:** spherical images. Both in taking position (see Fig. 12.8, p. 469). Epipolar plane $\mathcal{E}(X)$ through $O'O''X$, with the epipoles e' and e'' as images of the other projection centre; the straight or circular epipolar lines $l'(X)$ and $l''(X)$ which are the intersections of the epipolar plane $\mathcal{E}(X)$ and the image planes $\mathcal{D}'$ and $\mathcal{D}''$ or image spheres $\mathcal{S}'$ and $\mathcal{S}''$. Additional scene points, e.g., point $\mathcal{V}$, induce additional epipolar planes, building a pencil of planes with the base line $\mathcal{B} = O' \wedge O''$ as axis. Therefore, the epipolar lines also form a pencil of lines with the epipoles as carrier. Observe, χ' does not allow the inference of the position of X on the projecting line. Point $\mathcal{U}$ mapped to χ' , however, has a different image u'' . Both the point X and the point χ' induce the epipolar line $l''(X) = l''(\chi') = e'' \wedge \chi''$ in the other image

1. The *epipolar axis*

$$\mathcal{B} = O' \wedge O'' \quad (13.61)$$

is identical to the line through the two projection centres and contains the base line.⁵

2. The *epipolar plane*

$$\mathcal{E}(X) = O' \wedge O'' \wedge X \quad (13.62)$$

depends on the fixed projection centres O' and O'' and on the object point X . The epipolar planes build a pencil of planes with the epipolar axis as the common line. The epipolar planes therefore pass through the epipoles.

3. The *epipoles*

$$e' = \mathcal{P}'(O'') \quad e'' = \mathcal{P}''(O') \quad (13.63)$$

are the images of the other projection centres using the projections $\mathcal{P}'$ and $\mathcal{P}''$. The vectors $O'e'$ and $O''e''$ thus provide the direction to the other projection centres.

For perspective cameras the epipoles are the intersection points of the epipolar axis and the image planes $\mathcal{D}'$ and $\mathcal{D}''$

$$e' = (O' \wedge O'') \cap \mathcal{D}' \quad e'' = (O' \wedge O'') \cap \mathcal{D}'' . \quad (13.64)$$

For spherical cameras the epipoles are the intersection of the basis with the two image spheres $\mathcal{S}'$ and $\mathcal{S}''$.

4. The *epipolar lines*

$$l'(X) = \mathcal{P}'(O'' \wedge X) \quad l''(X) = \mathcal{P}''(O' \wedge X) \quad (13.65)$$

⁵ The base line $\mathcal{B}$ is not to be confused with the plane represented by the second row of a projection matrix (see Sect. 12.1.3.8, p. 473).

are the images of the rays $O' \wedge X$ and $O'' \wedge X$ in the other image, respectively.

For perspective and for spherical cameras they are the intersections of the epipolar plane with the image planes and spheres. They depend on the point X , e.g.,

$$\ell'(X) = \mathcal{E}(X) \cap \mathcal{D}' \quad \ell''(X) = \mathcal{E}(X) \cap \mathcal{D}''. \quad (13.66)$$

The projection centres O' and O'' , the object point X , the epipolar lines $\ell'(X)$ and $\ell''(X)$, and the two image points χ' and χ'' lie in the same epipolar plane.

In this case the prediction of χ'' can be solved easily: The epipolar plane $\mathcal{E}(\chi')$ is spanned by the three given points, the two projection centres O' and O'' , and the image point χ' ; its intersection with the other image plane yields the epipolar line $\ell''(\chi')$ on which the predicted point χ'' must lie.

These entities can easily be determined algebraically using the projection matrices or the fundamental matrix:

1. The epipolar axis has the direction of

$$\mathbf{b} = \mathbf{B} = \mathbf{Z}'' - \mathbf{Z}'. \quad (13.67)$$

We interpret it either as the vector $\mathbf{B}$ between the two projection centres or as its direction $\mathbf{b}$.

2. The epipolar lines are the projections of the projection lines $\mathcal{L}_{x'}$ and $\mathcal{L}_{x''}$ into the other images,

$$\mathbf{l}'(\mathbf{x}'') = \mathbf{Q}' \overline{\mathbf{Q}''}^T \mathbf{x}'', \quad \mathbf{l}''(\mathbf{x}') = \mathbf{Q}'' \overline{\mathbf{Q}'}^T \mathbf{x}', \quad (13.68)$$

using the equation $\mathbf{L}_{x'} = \overline{\mathbf{Q}'}^T \mathbf{x}'$ for the projection line (12.85), p. 483 with the projection $\mathbf{l}'' = \mathbf{Q}'' \mathbf{L}$ for object lines (12.72), p. 480. The epipolar lines ℓ' and ℓ'' are characterized by the constraint that corresponding points χ' and χ'' have to fulfil the coplanarity constraint $\mathbf{x}'^T \mathbf{F} \mathbf{x}'' = 0$ and the points have to lie on the lines, thus $\chi' \in \ell'$ and $\chi'' \in \ell''$. This is valid if

$$\mathbf{l}'(\mathbf{x}') = \mathbf{F} \mathbf{x}'', \quad \mathbf{l}''(\mathbf{x}') = \mathbf{F}^T \mathbf{x}', \quad (13.69)$$

since then $\mathbf{x}'^T \mathbf{l}' = 0$ and $\mathbf{x}''^T \mathbf{l}'' = 0$. The second equation is obviously the prediction line for χ'' in the second image if χ' is given in the first image.

Equation (13.69) is a remarkable transformation: points of one image are transformed into lines in the other image. This is one example of a broader group of dualizing transformations in projective geometry, known as projective correlation (see [Semple and Kneebone \(1998\)](#); [Tschupik and Hohenberg \(1972\)](#); [Niini \(2000\)](#) and Sect. 6.6, p. 282). Since the matrix $\mathbf{F}$ is singular, the mapping is also called a *singular correlation*. The notion has nothing in common with a statistical correlation, however. The relation

$$\mathbf{F} = \mathbf{Q}' \overline{\mathbf{Q}''}^T \quad (13.70)$$

can be used to prove (13.15) in a different way.

Equation (13.69) yields *oriented* epipolar lines if the fundamental matrix is determined from proper projection matrices from one of the equations in Sect. 13.2.2.2, p. 553. If the fundamental matrix is estimated from corresponding points, the sign needs to be adapted. A similar line of thought is valid also for the essential matrix. We discuss this in the context of deriving the base vector and the rotation from an estimated essential matrix (see Sect. 13.3.3, p. 581).

If the projection is *not distortion-free*, thus not straight line-preserving, we can first determine the projection line and project it into the other image by sampling the space line by a few points, yielding a *polygonal approximation of the curved epipolar line*.

epipolar lines ℓ' and ℓ''

fundamental matrix
as singular
correlation

3. The epipoles can be determined either directly from

$$\mathbf{e}' = \mathbf{P}'\mathbf{Z}'' \quad \text{and} \quad \mathbf{e}'' = \mathbf{P}''\mathbf{Z}' \quad (13.71)$$

or using the camera planes in the projection matrices, e.g., $\mathbf{P}' = [\mathbf{A}_1, \mathbf{B}_1, \mathbf{C}_1]^\top$ from

$$\mathbf{e}' = \begin{bmatrix} |\mathbf{A}_2, \mathbf{B}_2, \mathbf{C}_2, \mathbf{A}_1| \\ |\mathbf{A}_2, \mathbf{B}_2, \mathbf{C}_2, \mathbf{B}_1| \\ |\mathbf{A}_2, \mathbf{B}_2, \mathbf{C}_2, \mathbf{C}_1| \end{bmatrix} \quad \mathbf{e}'' = \begin{bmatrix} |\mathbf{A}_1, \mathbf{B}_1, \mathbf{C}_1, \mathbf{A}_2| \\ |\mathbf{A}_1, \mathbf{B}_1, \mathbf{C}_1, \mathbf{B}_2| \\ |\mathbf{A}_1, \mathbf{B}_1, \mathbf{C}_1, \mathbf{C}_2| \end{bmatrix}. \quad (13.72)$$

The second expressions result from the fact that the projection centres are the intersections of the camera planes, thus $\mathbf{Z}'' = -\mathbf{A}_2 \cap \mathbf{B}_2 \cap \mathbf{C}_2$ and $\mathbf{A}_1^\top \mathbf{Z}'' = -|\mathbf{A}_1, \mathbf{A}_2, \mathbf{B}_2, \mathbf{C}_2| = |\mathbf{A}_2, \mathbf{B}_2, \mathbf{C}_2, \mathbf{A}_1|$, using the dual of (5.117), p. 227. If the projection matrices, $\mathbf{P}'$ and $\mathbf{P}''$, are proper, both equations (13.71) and (13.72) yield oriented vectors for the epipoles. The epipole of an image, however, is incident with all its epipolar lines. Therefore, the epipoles $\mathbf{e}'$ and $\mathbf{e}''$ are the left and right eigenvectors of $\mathbf{F}$, respectively,

$$\mathbf{e}'^\top \mathbf{F} = \mathbf{0}^\top \quad \text{and} \quad \mathbf{F} \mathbf{e}'' = \mathbf{0}, \quad (13.73)$$

since for all $\mathbf{I}'$ we have $\mathbf{e}'^\top \mathbf{I}' = 0$ and for all $\mathbf{I}''$ we have $\mathbf{e}''^\top \mathbf{I}'' = 0$. Observe, calculating the epipoles from (13.73) does *not* necessarily yield oriented vectors $\mathbf{e}'$ and $\mathbf{e}''$.

The relations for *calibrated cameras* are similar, namely replacing the entities for perspective cameras by those for spherical cameras, i.e., the homogeneous vectors of the image points are replaced by the ray directions

$$\mathbf{x}' \rightarrow {}^c\mathbf{x}' = -\text{sign}(c')(\mathbf{K}')^{-1}\mathbf{x}', \quad \mathbf{e}' \rightarrow {}^c\mathbf{e}' = -\text{sign}(c')(\mathbf{K}')^{-1}\mathbf{e}'. \quad (13.74)$$

Observe, in each camera the vectors ${}^c\mathbf{e}'$ and ${}^c\mathbf{e}''$ provide the directions to the other projection centre.

Similarly, we need to replace the expressions for image lines, due to $((\mathbf{K}')^{-1})^\flat = \mathbf{K}'^\top/|\mathbf{K}'|$ and assuming $|\mathbf{K}'| > 0$,

$$\mathbf{I}' \rightarrow {}^c\mathbf{I}' = \mathbf{K}'^\top \mathbf{I}', \quad (13.75)$$

and for projection matrices,

$$\mathbf{P}' \rightarrow {}^c\mathbf{P}' = R'[I_3 | -\mathbf{Z}'], \quad \mathbf{Q}' \rightarrow {}^c\mathbf{Q}' = R'[-S(\mathbf{Z}') | I_3]; \quad (13.76)$$

see (12.17), p. 468 and (12.76), p. 481. For example, we obtain the expression for the essential matrix from (13.70), p. 564,

$$\mathbf{E} = {}^c\mathbf{Q}' {}^c\overline{\mathbf{Q}'}^\top = R'S(\mathbf{Z}'' - \mathbf{Z}')R'^\top, \quad (13.77)$$

replacing the fundamental matrix.

Remark: If the signs are not of interest, the factor $1/|\mathbf{K}'|$ can be omitted in (13.75) and (13.76), see Sect. 6.2.4, p. 258. $\diamond$

13.2.6 Generating Normalized Stereo Pairs

Image pairs can be viewed stereoscopically if certain preconditions are fulfilled; one of them is that the y -parallaxes are zero for all points. In addition, exploiting the epipolar geometry of an image pair is computationally more efficient if the geometry follows the normal case, see (Sect. 13.2.4). This is usually approximately fulfilled by aerial images for a larger image area but never for arbitrary image pairs.

epipoles $\mathbf{e}'$ and $\mathbf{e}''$

epipolar geometry
for calibrated
cameras

normalized image
pair

However, one may transform a given image pair into the normal case. A general solution uses as new epipolar lines the intersection of a cylinder (where the basis is its axis) with a pencil of planes around the basis and exploits the orientation of corresponding epipolar lines (Pollefeys et al., 1999). If the two viewing directions are not too close to the basis, an alternative is to keep the two bundles of rays and intersect them with a common plane parallel to the base vector such that the normal of the plane is as close as possible to the two viewing directions.

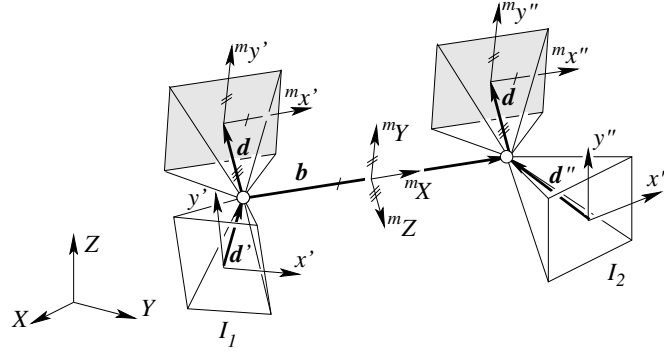


Fig. 13.7 Normalizing an image pair I_1, I_2 : the normalized stereo images, shaded gray, are chosen such that corresponding image points have no y -parallax. Thus the normalized images have a common calibration and rotation matrix, the common viewing direction $\mathbf{d}$, which is the average of the two viewing directions $\mathbf{d}_i$, rotated such that it is perpendicular to the basis. The x axes of the normalized images are parallel to the base vector $\mathbf{b}$. The common principal distance c is chosen to be negative, leading to normalized stereo images in viewing position

Now, the two bundles are kept and just the two image planes are replaced. The relations between the given image coordinates $\mathbf{x}'$ and $\mathbf{x}''$ and the transformed image coordinates ${}^m\mathbf{x}'$ and ${}^m\mathbf{x}''$ are two homographies,

$$\mathbf{x}' = \mathbf{H}'_m {}^m\mathbf{x}' \quad \mathbf{x}'' = \mathbf{H}''_m {}^m\mathbf{x}'' . \quad (13.78)$$

They may be used to digitally transform the two given images by indirect resampling. In the simplest case of ideal cameras we just have to rotate the image planes around the two projection centres.

The problem can be formalized as follows: Given are the two projection matrices $\mathbf{P}'$ and $\mathbf{P}''$, with the possibly different calibration matrices $\mathbf{K}'$ and $\mathbf{K}''$, the rotation matrices $\mathbf{R}'$ and $\mathbf{R}''$, and the two projection centres $\mathbf{Z}'$ and $\mathbf{Z}''$, leading to the projections

$$\mathbf{x}' = \mathbf{P}'\mathbf{X} = \mathbf{K}'\mathbf{R}'[\mathbf{I} - \mathbf{Z}']\mathbf{X} \quad \mathbf{x}'' = \mathbf{P}''\mathbf{X} = \mathbf{K}''\mathbf{R}''[\mathbf{I} - \mathbf{Z}'']\mathbf{X} . \quad (13.79)$$

The goal is to find the two homographies $\mathbf{H}'_m$ and $\mathbf{H}''_m$.

The idea is to achieve two new projections,

$${}^m\mathbf{x}' = {}^m\mathbf{P}'\mathbf{X} = \mathbf{K}\mathbf{R}[\mathbf{I} - \mathbf{Z}']\mathbf{X} \quad {}^m\mathbf{x}'' = {}^m\mathbf{P}''\mathbf{X} = \mathbf{K}\mathbf{R}[\mathbf{I} - \mathbf{Z}'']\mathbf{X} , \quad (13.80)$$

with the common calibration matrix $\mathbf{K}$ and common rotation matrix $\mathbf{R}$. The common calibration and rotation matrices guarantee the image content is different only with respect to x -parallaxes; the y -parallaxes are zero.

From (13.79) and (13.80) we obtain the homography matrices

$$\mathbf{H}'_m = \mathbf{K}'\mathbf{R}'\mathbf{R}^\top\mathbf{K}^{-1} \quad \mathbf{H}''_m = \mathbf{K}''\mathbf{R}''\mathbf{R}^\top\mathbf{K}^{-1} \quad (13.81)$$

or the inverse homographies

$${}^m\mathbf{x}' = \mathbf{K}\mathbf{R}\mathbf{R}'^\top\mathbf{K}'^{-1}\mathbf{x}' \quad {}^m\mathbf{x}'' = \mathbf{K}\mathbf{R}\mathbf{R}''^\top\mathbf{K}''^{-1}\mathbf{x}'' , \quad (13.82)$$

useful for digital image rectification. The homography matrices $\mathbf{H}'_m$ and $\mathbf{H}''_m$ are conjugate rotations, as each is similar to a rotation matrix (see Sect. 6.5.2, p. 281).

The common calibration matrix can be chosen to be $\mathbf{K} = \text{Diag}([c, c, 1])$. The sign of c should be chosen negatively in order to obtain the two transformed images in viewing position.

The rotation matrix $\mathbf{R}$ is chosen such that the two following conditions are fulfilled:

1. The ${}^m x'$ and the ${}^m x''$ axes of the transformed image coordinates are parallel to the base vector. This is identical to requiring the X axes of the two camera coordinate systems to be parallel to the base vector. Therefore the first row $\mathbf{r}_1^T$ of the rotation matrix is the normalized base vector $\mathbf{r}_1 = \mathbf{N}(\mathbf{B})$.
2. The viewing directions (the Z -axes of the camera coordinate systems) are orthogonal to the base vector and are as close as possible to the original viewing directions. One choice is to use the average viewing direction

$$\mathbf{d}^* = \mathbf{N} \left(\frac{\mathbf{d}'}{|\mathbf{d}'|} + \frac{\mathbf{d}''}{|\mathbf{d}''|} \right), \quad (13.83)$$

with (12.47), p. 475

$$\mathbf{d}' = -[p'_{31}, p'_{32}, p'_{33}]^T |\mathbf{A}'| \quad \mathbf{d}'' = -[p''_{31}, p''_{32}, p''_{33}]^T |\mathbf{A}''|. \quad (13.84)$$

and require the common viewing direction $\mathbf{d}$ to lie in the plane spanned by $\mathbf{b}$ and $\mathbf{d}^*$, cf. Fig. 13.7.

3. the ${}^m y'$ and the ${}^m y''$ axes of the transformed image coordinates are perpendicular to the base vector, thus the y -axes of the two camera coordinate systems are completing the Cartesian coordinate systems.

The rotation matrix therefore is

$$\mathbf{R} = [\mathbf{N}(\mathbf{b}), \mathbf{N}(\mathbf{b} \times \mathbf{d}^*), \mathbf{N}(\mathbf{b} \times (\mathbf{b} \times \mathbf{d}^*))]^T. \quad (13.85)$$

The common viewing direction then is $-\mathbf{N}(\mathbf{b} \times (\mathbf{b} \times \mathbf{d}^*))$.

13.2.7 Homography and Homology Induced by a Plane in Object Space

As a straight line-preserving mapping between two planes is a homography and the concatenation of two homographies is again a homography, the mapping between two images of the same plane in object space is a homography. The eight parameters of this homography can be described as a function of the five parameters of the relative orientation of two images of calibrated cameras and the three parameters of the plane in object space.

We start from the two projection matrices

$${}^c \mathbf{P}' := [I_3 | 0] = \mathbf{K}'^{-1} \mathbf{P}', \quad {}^c \mathbf{P}'' := \mathbf{R}[I | -\mathbf{B}] = \mathbf{K}''^{-1} \mathbf{P}'' \quad (13.86)$$

and a plane $\mathbf{A} = [\mathbf{A}_h^T, A_0]^T$. Then the mapping from the first to the second image is given by the homography, see Fig. 13.8,

*plane induced
homography*

$${}^c \mathbf{x}' = \mathbf{H} {}^c \mathbf{x}'' \quad \text{with} \quad \mathbf{H} = \mathbf{R} + \frac{\mathbf{T} \mathbf{A}_h^T}{A_0} \quad (13.87)$$

and

$$\mathbf{T} = \mathbf{R} \mathbf{B} \quad {}^c \mathbf{x}' = \mathbf{K}'^{-1} \mathbf{x}' \quad {}^c \mathbf{x}'' = \mathbf{K}''^{-1} \mathbf{x}'' . \quad (13.88)$$

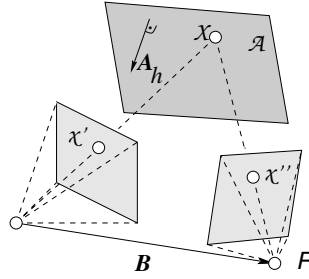


Fig. 13.8 Plane induced homography H between two images, which can be used for relative orientation with ≥ 4 points (see Sect. 13.3.2.5, p. 577). The homography ${}^c\mathbf{x}' = H^c\mathbf{x}''$ may be determined from ≥ 4 corresponding points $(\mathbf{x}'_i, \mathbf{x}''_i)$. It depends on the normal $\mathbf{A}_h$ of plane $\mathbf{A}$, the rotation matrix R of the second image with respect to the coordinate system of the first, and the base vector $\mathbf{B}$ via $\mathbf{T} = R\mathbf{B}$

Proof: The 3D coordinates of the 3D point $\mathbf{X} = [\mathbf{X}^T, 1]^T$ in the two camera coordinate systems are

$${}^1\mathbf{X} = \mathbf{X} \quad \text{and} \quad {}^2\mathbf{X} = R(\mathbf{X} - \mathbf{B}) = R\mathbf{X} - \mathbf{T}. \quad (13.89)$$

If we use the relation $\mathbf{A}^T\mathbf{X} = \mathbf{A}_h^T\mathbf{X} + A_0 = 0$, thus $-\mathbf{A}_h^T\mathbf{X}/A_0 = 1$ for all points $\mathbf{X}$ on the plane, we obtain

$${}^2\mathbf{X} = R\mathbf{X} - \frac{-\mathbf{T}\mathbf{A}_h^T\mathbf{X}}{A_0} = \left(R + \frac{\mathbf{T}\mathbf{A}_h^T}{A_0}\right) {}^1\mathbf{X},$$

from which (13.87) follows due to ${}^c\mathbf{x}' = {}^1\mathbf{X}$ and ${}^c\mathbf{x}'' = {}^2\mathbf{X}$. $\diamond$

If the rotation matrix is a unit matrix, the two images are related by a translation only, and the mapping between the images is a *homology*. It is given by

$$H = I_3 + \frac{\mathbf{T}\mathbf{A}_h^T}{A_0} \quad (13.90)$$

and only depends on five parameters, namely the direction vectors $\mathbf{T}$ and $\mathbf{A}_h$ and the parameter A_0 . Then two singular values of the matrix H are identical.

13.3 Relative Orientation of the Image Pair

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In the following section we discuss the determination of the relative orientation of two images from image point measurements, without control points or lines. This problem occurs

1. when observing a possibly moving object simultaneously with two cameras,
2. when observing a static object at two time instances with a moving camera (called structure from motion in the computer vision literature), and
3. when observing a moving object from a static camera, neglecting the stable background.

Only corresponding points can be used, as corresponding straight lines do not give a constraint on the relative orientation. The determination of the relative orientation of two images is based here on the coplanarity equation. The procedure is nonlinear in

plan induced
homology for pure
translation